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| <b>30</b> | <b>C</b> | <p>If a 10 g sample of ancient wood causes a count rate of 50 counts per minute, then a 5 g sample of the same wood would cause a count rate of 25 counts per minute.</p> <p>Thus for a fixed mass, the count rate of the ancient wood is <math>\frac{1}{4}</math> of the count rate of living wood. This means that two half-lives have passed and so, the age of the ancient wood = <math>2 \times 5600 = 11200</math> years.</p> |